

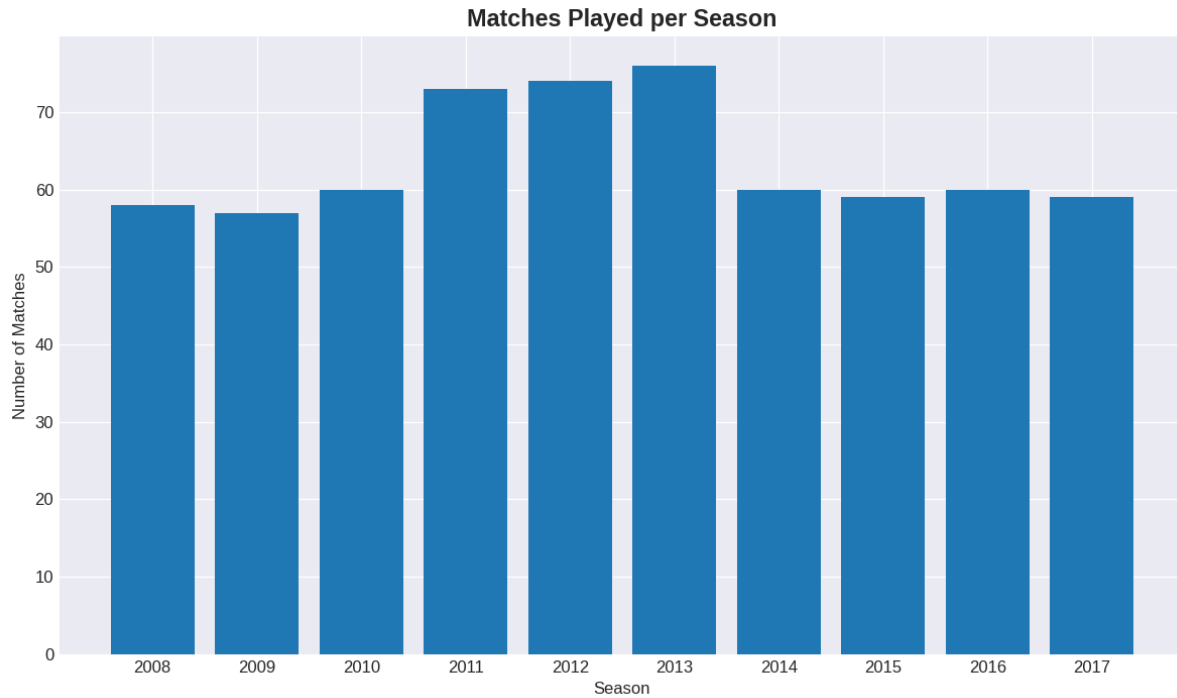
IPL Data Analysis

Insights Report | Seasons 2008 - 2017

This report summarizes key findings from an exploratory data analysis of Indian Premier League (IPL) match and ball-by-ball data covering 636 matches across 10 seasons. The analysis covers team performance, player performance, toss impact, venue trends, and scoring patterns.

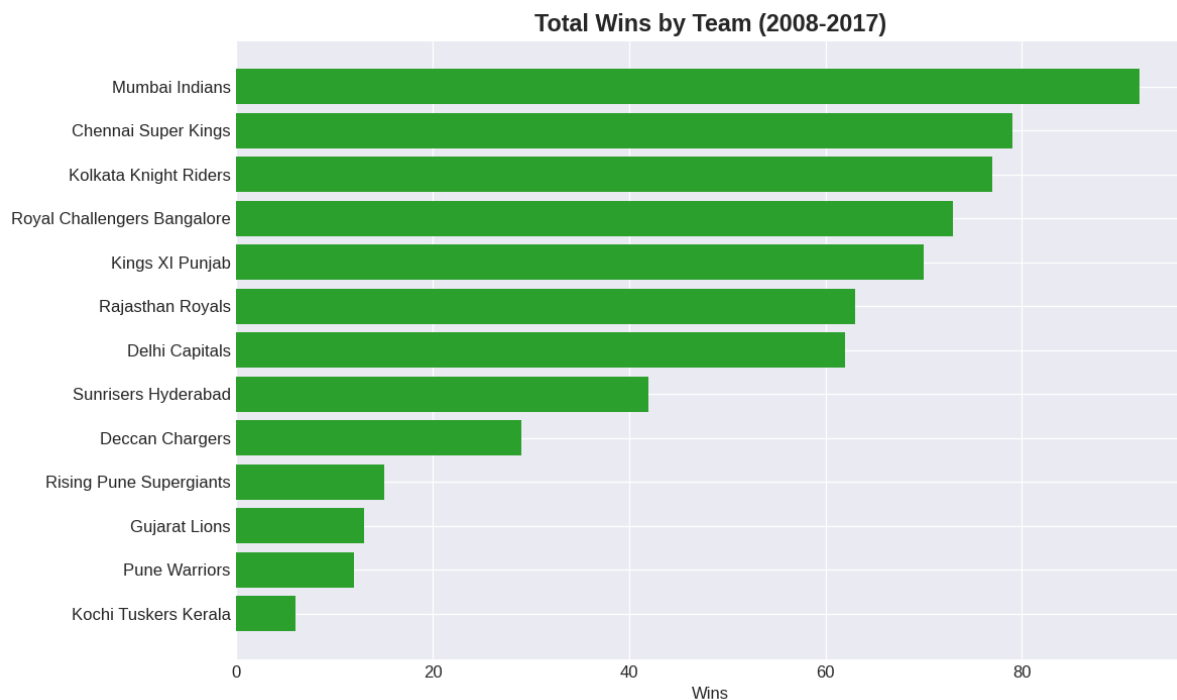
Metric	Value
Total Matches Analyzed	636
Seasons Covered	2008 - 2017 (10 seasons)
Total Deliveries (Balls)	179,078
Most Successful Team	Mumbai Indians (92 wins)
Top Run Scorer	V Kohli (5,434 runs)
Top Wicket Taker	SL Malinga (170 wickets)
Toss Winner Match Win Rate	51.1%

1. Matches Played Per Season



- Season count has stayed largely stable at 60 matches/season except 2011 & 2012 (larger 9-10 team format).
- Indicates a mature, standardized tournament structure since 2013.

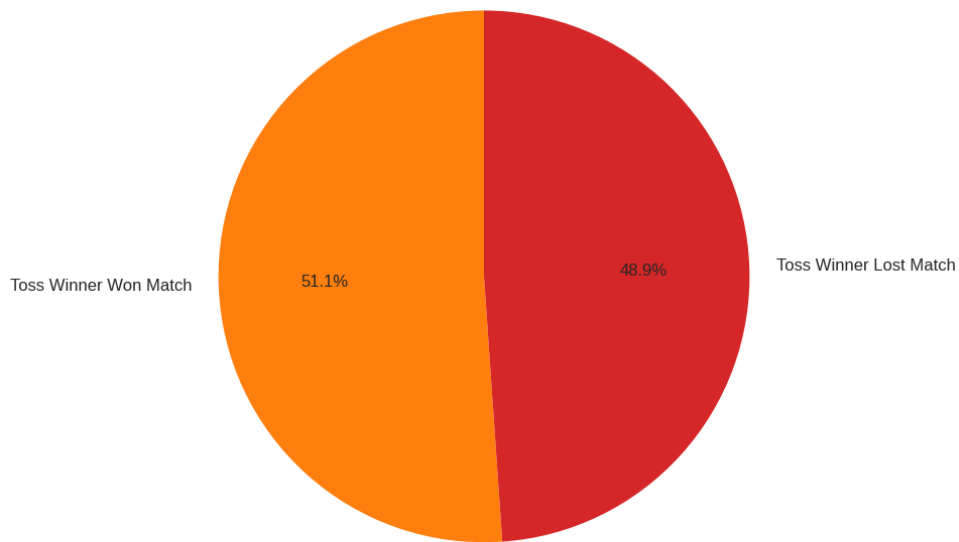
2. Most Successful Teams (Total Wins)



- Mumbai Indians lead with 92 wins, followed closely by Chennai Super Kings and Kolkata Knight Riders.
- The top 4 franchises account for a disproportionate share of total wins, showing consistent squad strength.

3. Toss Impact on Match Outcome

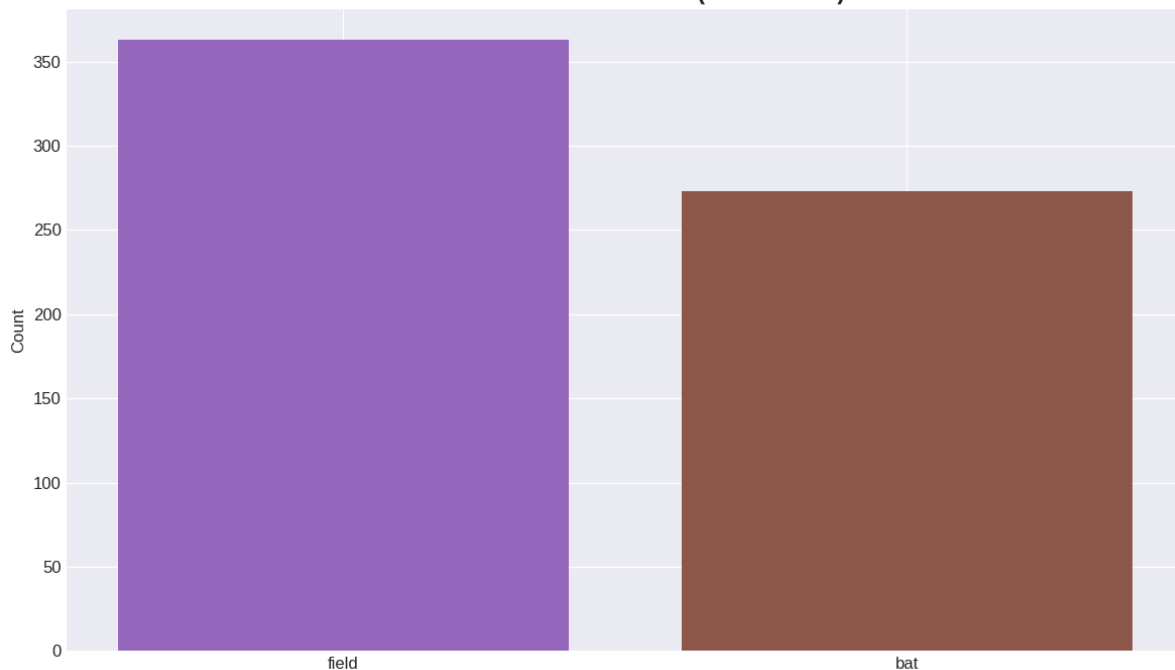
Does Winning the Toss Help Win the Match?



- Winning the toss translates to a match win only ~51% of the time - close to a coin flip.
- Toss advantage is commonly over-stated; on-field performance matters far more than the toss.

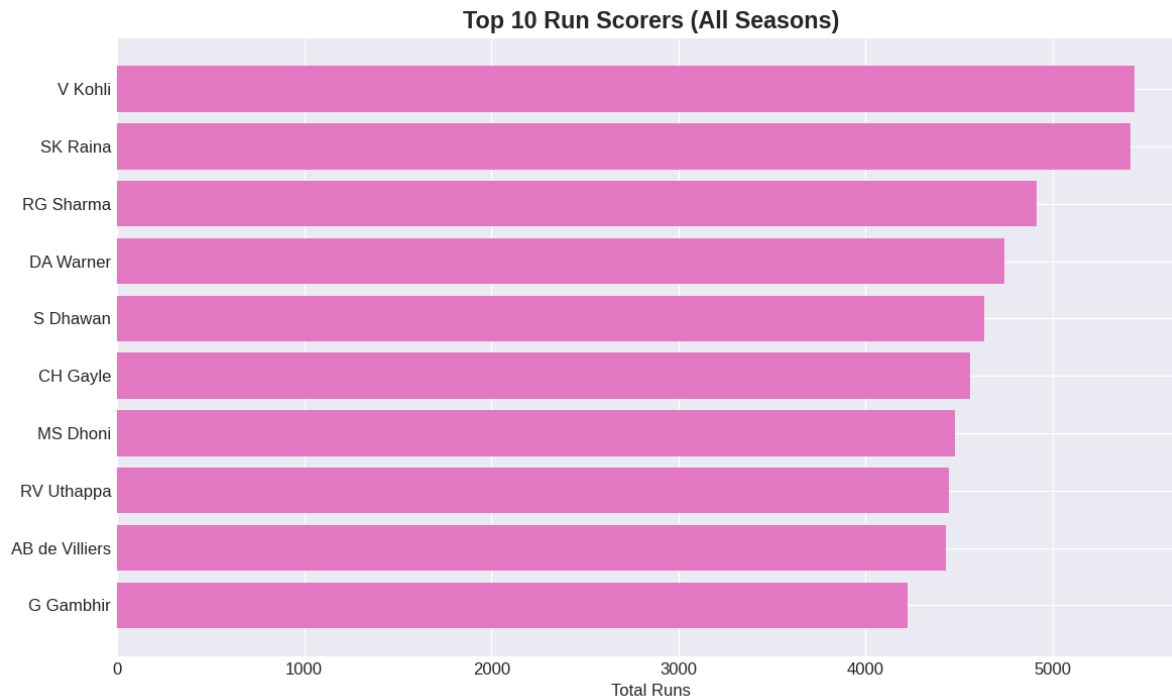
4. Toss Decision Preference

Toss Decision Preference (Bat vs Field)



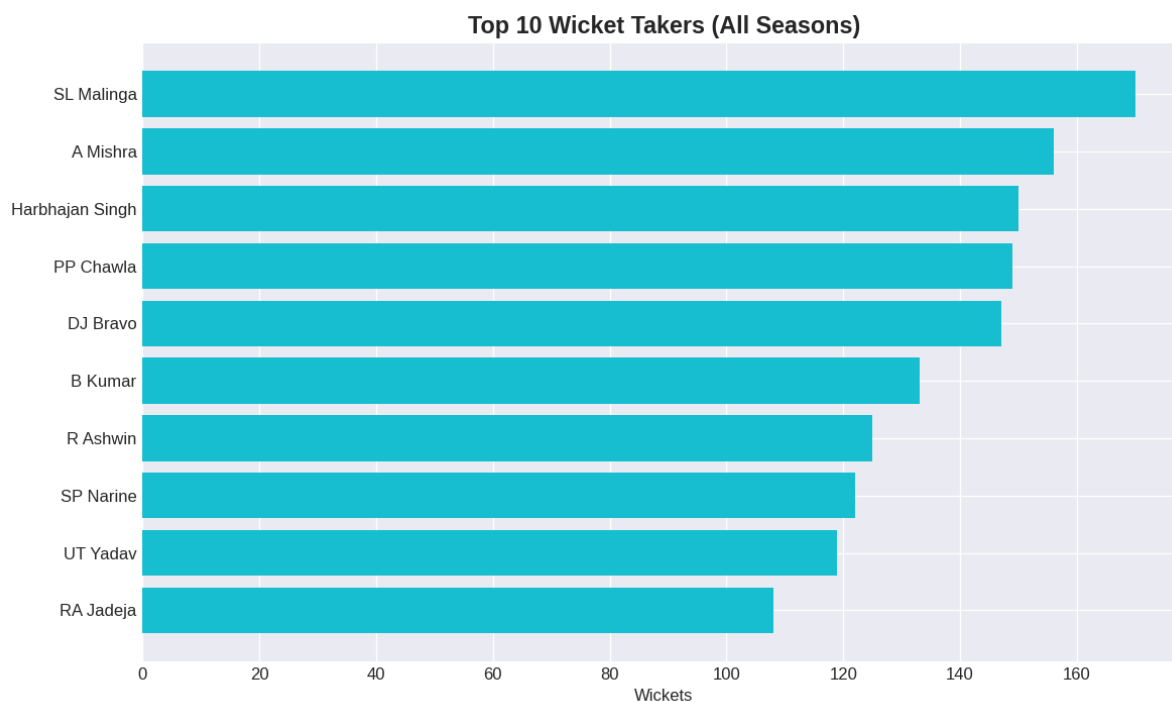
- Teams overwhelmingly prefer to field first after winning the toss.
- Reflects the strategic advantage of chasing under dew/lights in T20 cricket.

5. Top 10 Run Scorers



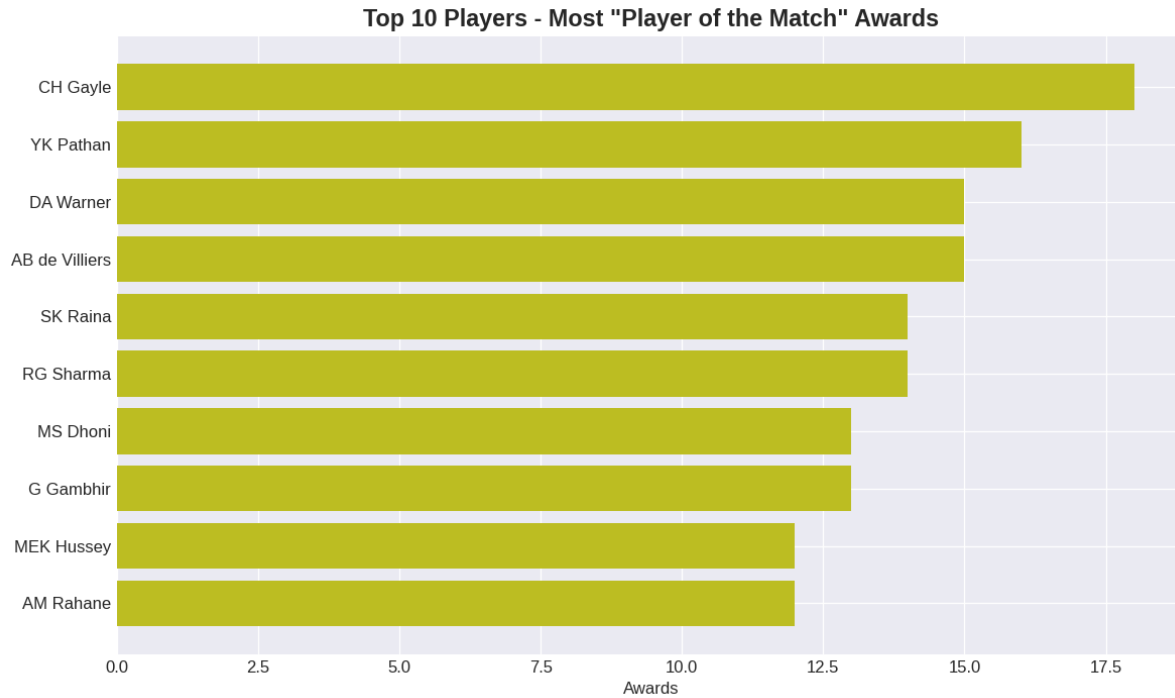
- Virat Kohli tops the all-time run charts, reflecting sustained batting excellence across seasons.
- Top run-scorers are dominated by top/middle-order batsmen from consistently strong franchises.

6. Top 10 Wicket Takers



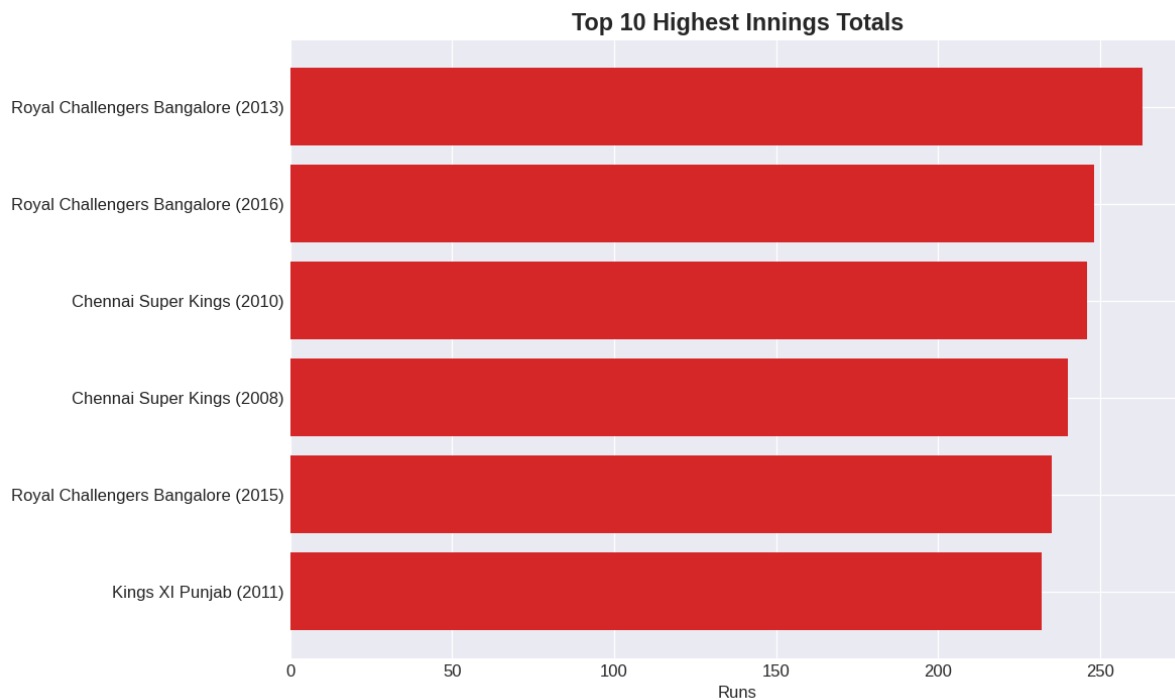
- SL Malinga leads all bowlers in total wickets, driven by his death-overs specialization.
- Spin and pace bowlers both feature in the top 10, showing balanced bowling success across styles.

7. Most 'Player of the Match' Awards



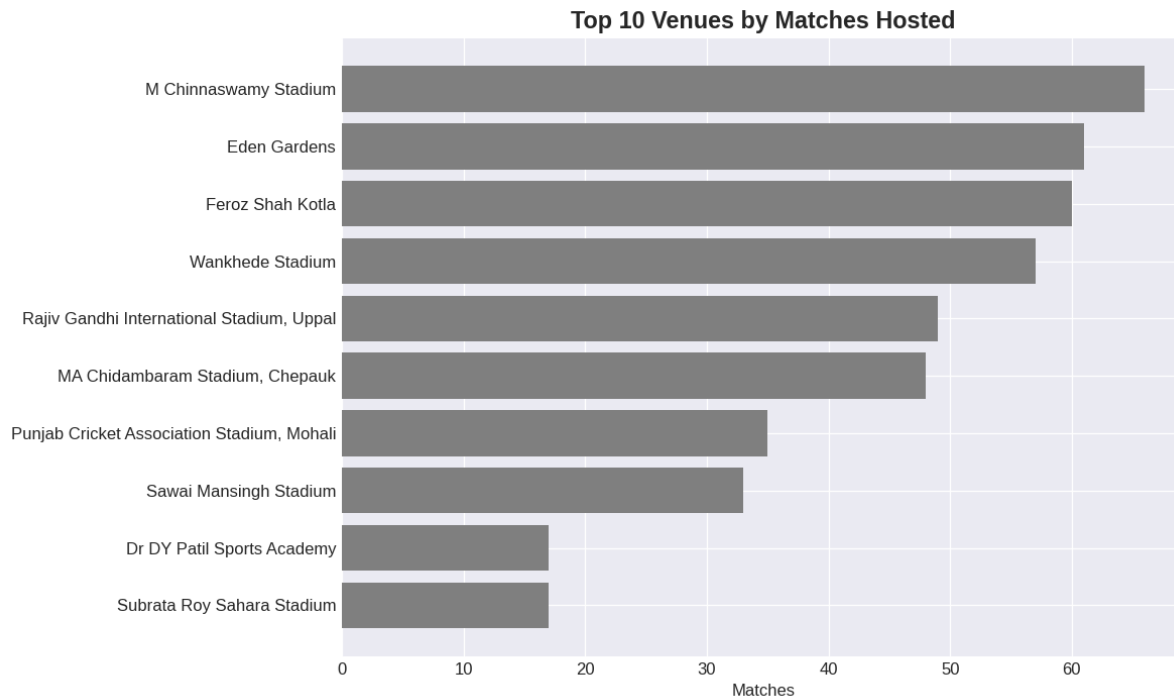
- Individual match-winning performances are concentrated among a small set of star all-rounders and batsmen.
- These players are high-value assets for match-defining moments, useful for team-building strategy.

8. Highest Innings Totals



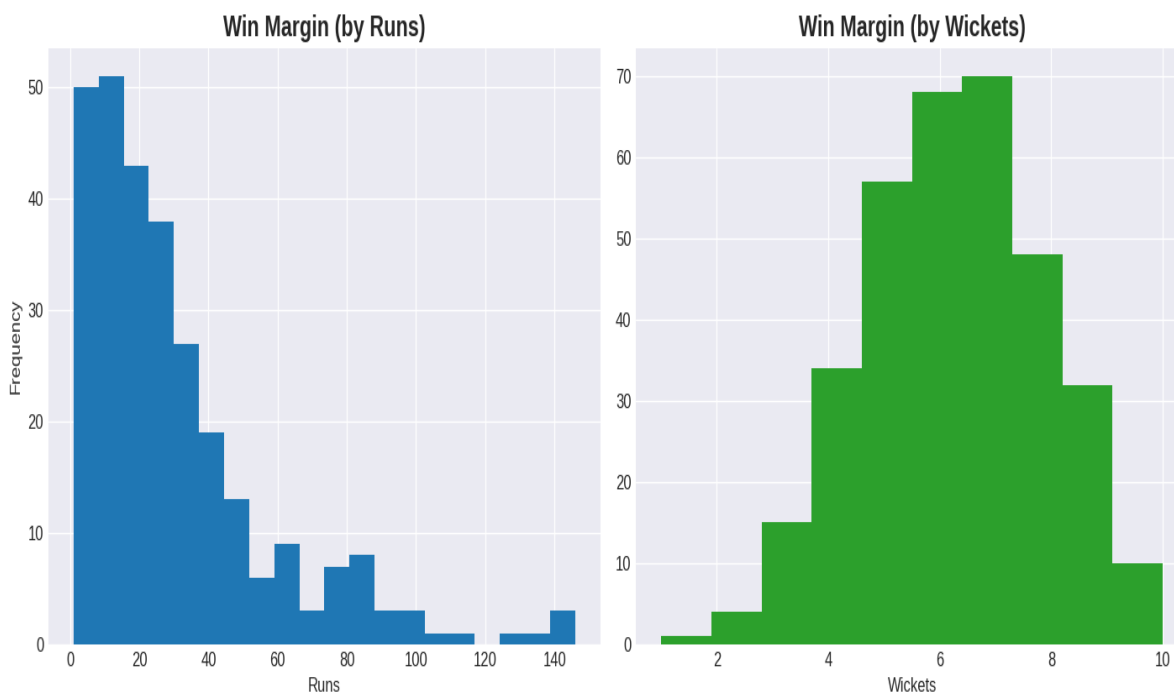
- The highest team totals recorded exceed 260 runs in a single innings.
- Explosive top-order batting and favorable batting-first conditions drive these record scores.

9. Top 10 Venues by Matches Hosted



- A handful of venues (major metro stadiums) host a disproportionately large share of matches.
- Venue familiarity may give home teams a slight tactical edge.

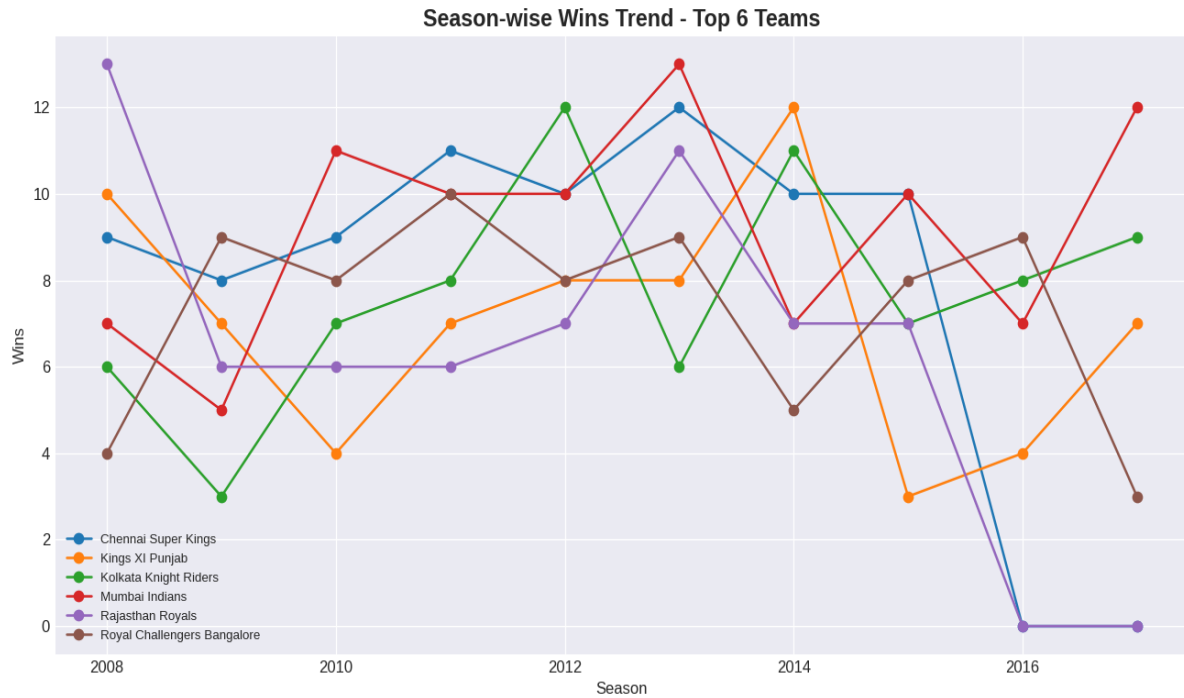
10. Win Margin Distribution



- Most run-based wins are in the 10-40 run range; blowout wins (80+ runs) are rare.

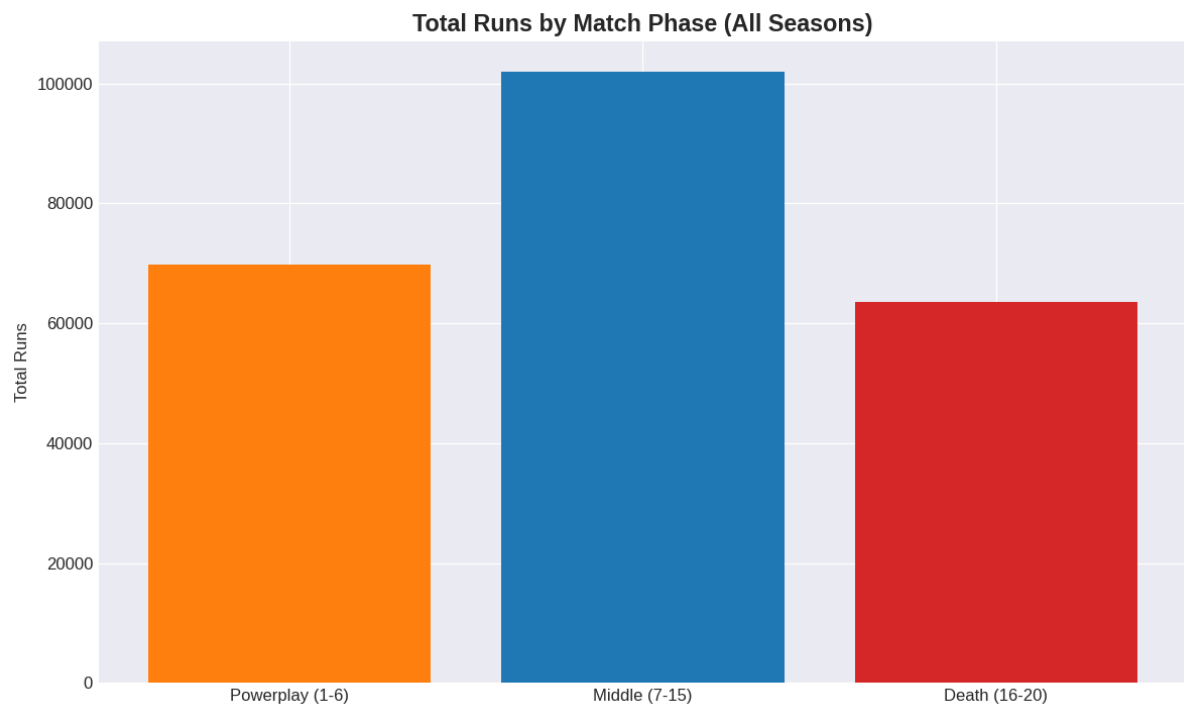
- Wicket-based wins cluster around 5-8 wickets remaining, indicating comfortable but not dominant chases.

11. Season-wise Wins Trend - Top 6 Teams



- Mumbai Indians and Chennai Super Kings show the most consistent top-tier performance across seasons.
- Other franchises show more volatile, boom-bust performance year to year.

12. Scoring Pattern by Match Phase



- Death overs (16-20) produce the highest run output, confirming aggressive 'slog overs' scoring.
- Powerplay scoring is lower in aggregate due to fielding restrictions balancing bat vs ball.

Conclusion & Recommendations

This analysis of 10 seasons of IPL data highlights that consistent franchise performance (Mumbai Indians, Chennai Super Kings) stems from stable core squads rather than one-off toss luck or venue advantage. Key takeaways for a data-driven decision-making context:

1. **Toss is not a reliable predictor** of match outcome (~51% win rate) - team composition and death-overs execution matter more.
2. **Death-overs specialists** (batting and bowling) disproportionately influence match results and should be prioritized in squad planning.
3. **Fielding first after toss** is now the dominant strategy across the league.
4. **Star players** with repeated Player-of-the-Match awards are reliable indicators of match-winning ability and franchise value.

This end-to-end analysis (Python notebook + Power BI dashboard + this report) is designed to demonstrate applied data analytics skills including data cleaning, aggregation, visualization, and business storytelling.